CLAIMS LEAKAGE DETECTION / LEVEL 1

Find the leakage before you pay.

ClaimIQ turns fragmented claim, policy, invoice, payment and document data into evidence-led financial reviews - so teams can spot preventable overpayments before they become losses.

50

synthetic claims

in the repeatable demo

14

planted scenarios

including a valid lookalike

1

batched AI review

optional per analysis run

Not another black-box fraud score.

ClaimIQ reconstructs the claim financial story, shows the source evidence behind each finding, and explains the amount at stake. Suspicious patterns stay reviewable; a person makes the final decision.

WHAT IT FINDS

Duplicates, policy-limit and exclusion issues, settlement errors, and missing evidence.

WHAT IT RETURNS

A counted-once financial view, claim-level proof, and an auditable review path.

From raw files to review-ready evidence

One traceable path from messy source data to a human-reviewed financial decision.

1

INGEST

CSV, Excel, JSON, XML, ZIP and PDF tables.

2

MAP

Suggest data roles and fields; let the handler review mappings.

3

DETECT

Apply policy and duplicate checks; calculate impact deterministically.

4

EXPLAIN

Show source rows, calculation steps and possible legitimate context.

5

RESOLVE

Confirm, correct or dismiss; preserve the action in the audit trail.

DETECTION COVERAGE

- Duplicate invoices, lines and payments
- Near-duplicates and repeated claim rows
- Coverage, room, sub-limit and exclusions
- Deductible / co-pay settlement recalculation
- Policy-period, waiting-period and document checks

BUILT FOR REVIEW

- Overlapping findings do not double-count a charge.
- Missing evidence is surfaced for clarification.
- LLM adds review text; deterministic rules own amounts.
- Authorised-role decisions, audit history and CSV export.
- Unreadable files and quarantined rows are made visible.

TECHNOLOGY Next.js + React interface / FastAPI + Python services / Pandas ingestion + deterministic rules / optional Gemini or OpenAI-compatible LLM

A clear boundary between calculation and language

The rules engine detects candidates and computes financial amounts. When configured, one batched LLM call can add evidence-review guidance, lookalike assessments and explanations; it cannot change the math. Without an LLM key, the deterministic workflow still runs. This is decision support - not automated claim adjudication.

02 / LIVE JUDGE DEMO

A focused 90-second walkthrough

Use the built-in synthetic sample to show the full experience - no external AI key or upload preparation required.

From the project root, after installing dependencies (see README).

TERMINAL 1 / BACKEND

```
cd claimsentry\backend
python -m uvicorn app.main:app --port 8000
```

API :8000

TERMINAL 2 / FRONTEND

```
cd claimsentry\frontend
npm run dev
```

WEB :3000

SHOW THE VALUE, NOT JUST THE DASHBOARD

1. Open http://localhost:3000 and choose "Load sample (with my rules)".
2. Review the leakage report: totals, findings, policy checks and legitimate lookalikes.
3. Open a finding to show source evidence and the step-by-step amount calculation.
4. Confirm, correct or dismiss a finding; show the updated total and audit trail.

WHAT THE SAMPLE PROVES

50 synthetic claims; 14 planted scenarios spanning duplicate invoices and payments, duplicate lines, policy limits, exclusions, settlement errors, missing documents and related-party patterns. A weekly physiotherapy lookalike is intentionally treated as legitimate. The test suite checks planted cases and that overlapping amounts are not double-counted.

The decision stays with the claims handler.

ClaimIQ prioritises evidence and transparent calculations. Data quality affects what can be assessed; missing or ambiguous support is surfaced for review. Optional LLM features require a configured provider key, while core detection and arithmetic remain deterministic.